with a great deal of confidence, leave them to be worried over by whatever post-human entities might inhabit the cosmos after we're gone. Vacuum decay is special in that it could technically happen at any moment, even if the chances of that are astronomically low. It also comes with a uniquely extreme, almost gratuitous finality.

In 1980, two theorists, Sidney Coleman and Frank De Luccia, calculated that a true vacuum bubble would contain not only a totally different (and lethal) arrangement of particle physics, but also a kind of space that is, by its nature, gravitationally unstable. Once the bubble formed, they explained, everything inside would collapse gravitationally within microseconds. Then they wrote:

> *This is disheartening. The possibility that we are living in a false vacuum has never been a cheering one to contemplate. Vacuum decay is the ultimate ecological catastrophe; in a new vacuum there are new constants of nature; after vacuum decay, not only is life as we know it impossible, so is chemistry as we know it. However, one could always draw stoic comfort from the possibility that perhaps in the course of time the new vacuum would sustain, if not life as we know it, at least some structures capable of knowing joy. This possibility has now been eliminated.*[fn10]

THE JOY OF NOT KNOWING

Of course, vacuum decay is, relatively speaking, a fairly new idea that incorporates so many kinds of extreme physics that it's entirely conceivable that our perspective on it will shift dramatically over the next few years. It may be that more detailed, rigorous calculations will give us different answers. These questions are difficult and complicated, and we still have a way to go before a consensus is reached.

If we conclude that our vacuum really is metastable, this may be incompatible with the theory of cosmic inflation. The quantum fluctuations during inflation, or the ambient heat afterward, seem like they should have been sufficient to trigger vacuum decay in the first moments of the cosmos, negating our very existence. Clearly, that didn't happen. Which suggests either we don't understand the early universe, or vacuum decay was never possible at all.